

VEDAVYAS ROKALABOINA

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Eligibility: Valid UK Right to Work (Graduate Visa)

PROFESSIONAL SUMMARY

MSc-qualified Mechanical Design Engineer with strong CAD, GD&T, and engineering-calculation skills, built through structural, aerodynamic, and systems design projects — including a 42,614 kg aircraft sizing study, a 1.9-million-element CFD validation to under 1% deviation, and a six-geometry Mach 1.8 nozzle comparison. Experienced in taking concepts through parametric 3D modelling in SolidWorks and CATIA V5 to fully tolerance 2D production and general arrangement drawings, and in producing engineering calculations and technical documentation to support project specification. Comfortable working as part of a design team to deliver systems on time, contributing to design reviews, and coordinating across disciplines under pressure, looking to bring this design and technical documentation background into a fast-paced engineering/manufacturing environment.

CORE SKILLS & TECHNOLOGIES

CAD Design: SolidWorks (advanced, multi-part assemblies), CATIA V5, AutoCAD, Geometric Dimensioning & Tolerancing (GD&T), 2D production and general arrangement drawing generation

Engineering Calculations & Analysis: MATLAB-based sizing and load calculations, ANSYS Fluent/CFX simulation, mesh convergence studies (grid-independence from 1M to 1.9M+ elements), Reynolds number and y+ wall-treatment calculation, FEA fundamentals, structural sizing fundamentals

Programming: MATLAB, Python

Documentation & Compliance: Engineering documentation, bills of material, validation records, technical report writing, tolerance stack-up awareness

Soft Skills: Problem-solving, cross-functional team collaboration, design review contribution, technical communication.

ENGINEERING WORK EXPERIENCE

Mechanical Design Engineer (Part-time, Remote) | Abhoday Bharat Projects Pvt Ltd Mar 2024 – Sep 2025

- **CAD & Drafting:** Worked remotely and part-time alongside full-time MSc study, supporting mechanical design and configuration of hardware components in SolidWorks and CATIA V5, converting specifications into manufacturing-ready 3D models and producing 2D technical drawings with full GD&T annotation.
- **Interference & Tolerance Checks:** Ran interference checks across assemblies and validated tolerance stack-ups before releasing 2D drawings for machining.
- **Cross-Functional Liaison:** Liaised with the electrical/controls team on packaging and space envelopes for integrated hardware, coordinating design intent across disciplines to agreed specifications.
- **Simulation Support:** Supported CFD and structural FEA analysis in ANSYS Fluent to model mechanical load reactions and track performance data against design targets.
- **Documentation:** Helped maintain engineering documentation, bills of material, and validation records to support the transition from prototype to production.
- **Design Corrections:** Supported root cause analysis on mechanical assemblies, contributing engineering calculations toward design corrections that improved one component's strength-to-weight margin by approximately 15%.

Design Team Member | ARDC, IARE Apr 2022 – Feb 2024

- **Fabrication & Testing:** Part of a multi-disciplinary student team that drafted, fabricated, and lab-tested a 200-inch wing structure assembly, including manual workshop tooling and physical model validation.
- **Design Calculations:** Developed design calculations and parametric models in XFLR5 and SolidWorks to run continuous performance loops, lifting baseline structural efficiency metrics by 20%.
- **Design Review Contribution:** Helped compile technical evidence and contributed to the project's design review presented before appraisal panels at the SAE International competition (USA), achieving a top-15 global ranking out of the full international field.

ENGINEERING PROJECTS & SIMULATION RESEARCH

Aerodynamics of a Bio-Inspired Blended Wing Body Apr 2025 – Sep 2025

- **Thermal-Fluid Simulation:** Ran a two-stage CFD validation in ANSYS Fluent (k-omega SST): a mesh convergence study progressing from 1 million to 2 million elements at a constant 4° angle of attack, converging on a grid-independent 1.9-million-element mesh at a Reynolds number of 4×10^6 .

- **Boundary Layer Resolution:** Calculated first-layer mesh height from a target y^+ value of 1 using boundary-layer wall-function relations and matched the 40 m/s free-stream velocity and 4° angle of attack of a published benchmark study to validate the drag coefficient to within under 1% deviation.
- **Inflation Layer Meshing:** Configured near-wall inflation layers for both the validation wing (22 layers, 10 power -5 m first-layer thickness, 1.22 growth rate, 3.6 mm maximum thickness) and the full BWB aircraft (20 layers, 8.2×10 power -6 m first-layer thickness, 1.2 growth rate, 3.24 mm maximum thickness) to ensure smooth near-wall-to-far-field mesh transition.
- **Design Verification:** Ran a parametric geometry sweep (winglet sweep angle 30° – 60°) at cruise condition, using pressure-contour and velocity-streamline analysis to verify attached flow and select the optimum configuration.
- **Reporting:** Wrote up the full validation methodology and results ^{2as} an academic dissertation, including post-processing scripts for result verification.

POV Business Jet Design, Performance Optimisation & Simulation

Sep 2024 – Mar 2025

- **CAD & Systems Integration:** Engineered an integrated multi-component 3D assembly in SolidWorks — wing, tail, fuselage, and retractable landing gear — from calculated sizing dimensions (28.092 m fuselage length, 29.64 m wingspan, 5.452 m root/1.36 m tip chord, 30° wing sweep), coordinating geometry across interfacing sub-systems and checking physical clearances.
- **Tail Sizing:** Sized vertical (21.31 square-meter area, 5.056 m span) and horizontal (15.24 square-meter area) tail surfaces using Raymer's tail volume coefficient method, calculating root chord and taper geometry for each surface.
- **Numerical Sizing:** Programmed MATLAB scripts to evaluate weight, load, and sizing matrices for a 4-crew aircraft, calculating a required 61 kN combined thrust from twin engines and a stall wing loading of 85.68 lb/square-feet, cross-checking results against published reference data.
- **Take-Off Performance:** Calculated a Take-Off Parameter (TOP) of 230 against a 5,860 ft (~1,800 m) target field length, using Raymer's TOP-vs-distance method to size take-off wing loading and thrust requirements.
- **Simulation & Validation:** Ran SolidWorks Flow Simulation for cruise aerodynamics and cross-checked flight characteristics against theoretical predictions using a flight simulator, iterating the design across three versions to refine performance.

Thrust Vector Control by Supersonic Jets

Jun 2023 – Apr 2024

- **Geometry & Specification Control:** Sized a converging-diverging nozzle from isentropic flow relations (380.13 square-millimeter inlet, 75 square-millimeter throat, 108 mm² exit, $A_e/A^* = 1.39$, 70 mm convergent/6.5 mm divergent length) and modelled six exit geometries in CATIA V5 for a design exit Mach number of 1.8.
- **Simulation & Reporting:** Matched mesh density across all six geometries (35,800–36,900 nodes; 197,800–204,200 elements) and boundary conditions (12 atm/288 K inlet, 1 atm/288 K ambient) in ANSYS ICEM CFD/CFX (k-omega SST, high-resolution advection, 100 iterations) for a fair comparative study, compiling results into a structured technical report.
- **Design Comparison:** Compared supersonic core length across geometries (circular 23, square 24, rectangular 21, elliptical 18, hexagonal 15, triangular 10) against a circular baseline to identify the closest-performing non-circular alternative, supporting informed geometry selection under design constraints.

EDUCATION

Master of Science (MSc) in Aeronautical Engineering

Sep 2024 – Sep 2025

University of South Wales, United Kingdom

Relevant modules: Flight Vehicle Systems Design, Computational Structural Analysis (Stress Awareness), Advanced Fluid Dynamics Modelling, Computational Simulation Techniques.

Bachelor of Technology (BTech) in Aeronautical Engineering

Dec 2020 – Apr 2024

Institute of Aeronautical Engineering, India

Relevant modules: Engineering Materials Technology, Structural Dynamics, Mechanical Vibration Analysis, Aerodynamics, Advanced CAD Modelling.

PERSONAL PROJECT

Aircraft Conceptual Sizing Studio

Live app: aircraftsizing.streamlitapp

- **Build & Methodology:** Built and deployed an interactive Python/Streamlit web application implementing a conceptual sizing methodology across 8 aircraft categories, computing gross weight, thrust, and wing/tail/landing-gear geometry from mission inputs.
- **Validation & Debugging:** Validated results against 12 real, named aircraft, achieving a worst-case deviation of 14% from published data; iteratively diagnosed and corrected sizing-model errors, recalibrating constants against published reference data.